Wireless Communication Technology

Wireless – allows transmission of information via ***Electromagnetic Waves***

Radio, Acoustic,
Infrared, Satellite

NO wires, cables,
conductors

Types of Wireless Communication Technologies

- Satellite Communication
- Wireless Networking (Wi-Fi, Wi-Max)
- Bluetooth
- ZigBee

Wireless Devices

Transistor
Walkie-Talkie
Wireless Phone
Remote Devices
Wireless Sensors
Video Game Consoles

Cordless Phone
Cellular Phones

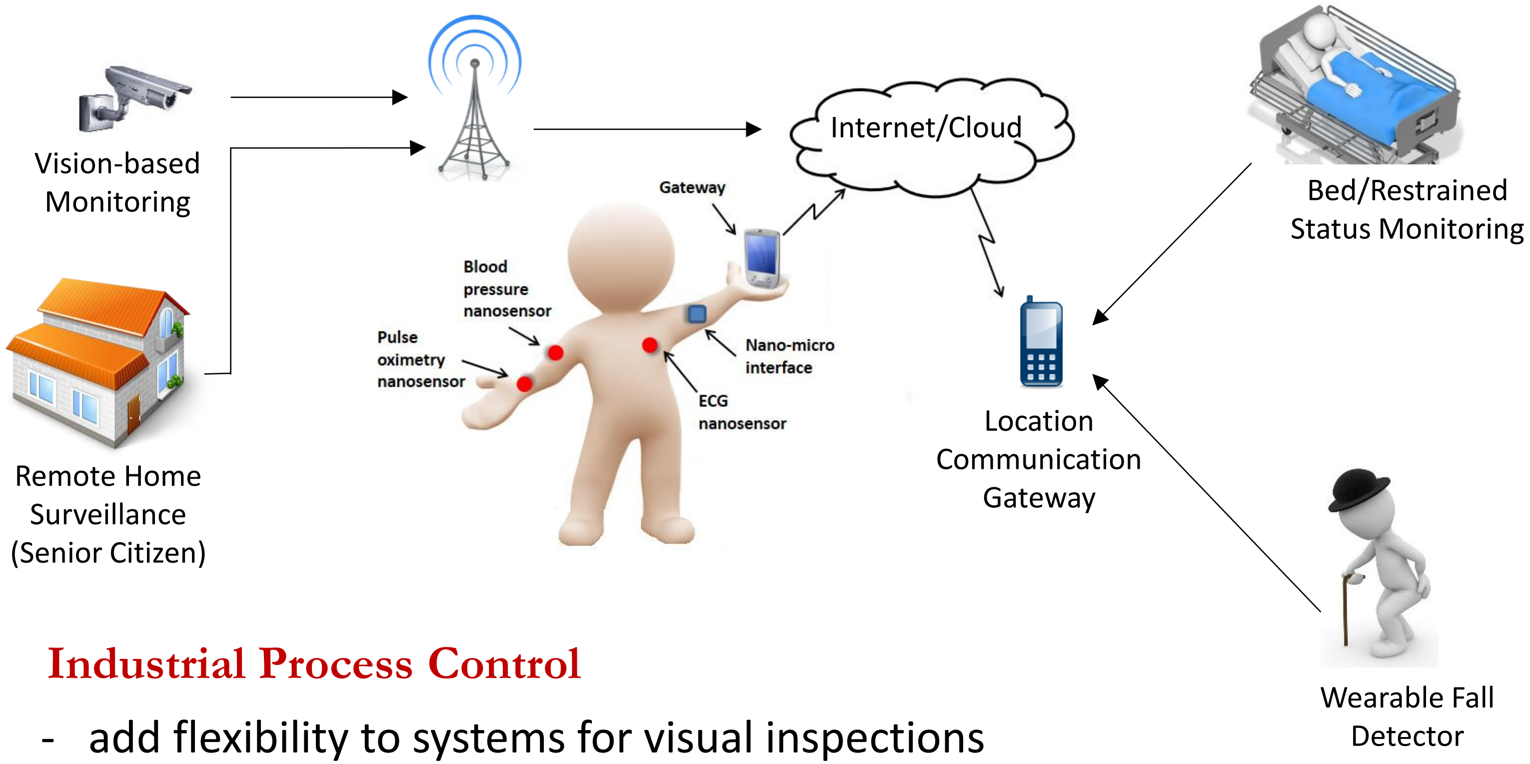
Radio Frequency Bands

Band Name	Frequency	Wavelength	Applications
Extremely Low Frequency (ELF)	30 to 300 Hz	10,000 to 1,000 Km	Powerline frequencies
Voice Frequency (VF)	300 to 3,000 Hz	1,000 to 100 Km	Telephone communications
Very Low Frequency (VLF)	3 to 30 KHz	100 to 10 Km	Marine communications
Low Frequency (LF)	30 to 300 KHz	10 to 1 Km	Marine communications
Medium Frequency (MF)	300 to 3,000 KHz	1,000 to 100 m*	AM broadcasting
High Frequency (HF)	3 to 30 MHz	100 to 10 m	Long-distance aircraft/ship communications
Very High Frequency (VHF)	30 to 300 MHz	10 to 1 m	FM broadcasting
Ultra High Frequency (UHF)	300 to 3,000 MHz	100 to 10 cm	Cellular telephone
Super High Frequency (SHF)	3 to 30 GHz	10 to 1 cm	Satellite communications, microwave links
Extremely High Frequency (EHF)	30 to 300 GHz	10 to 1 mm	Wireless local loop
Infrared	300 GHz to 400 THz	1 mm to 770 nm	Consumer electronics
Visible Light	400 THz to 900 THz	770 nm to 330 nm	Optical communications

Frequency bands in electromagnetic spectrum defined by ITU

Recent Trend Applications

- **Advanced Health Care Delivery**
  - patients carry medical sensors to monitor certain physical parameters
  - assist senior citizens remotely



- **Industrial Process Control**
  - add flexibility to systems for visual inspections
  - enhanced automated actions

Multimedia Applications

Information System

- Electronic Publishing
- Online Catalogs
- Patient Monitoring System
- Shopping Mall Navigation

Remote Representation

- Video Conferencing
- Distance Learning
- Remote Actions
- Robot Surveillance (nuclear reactor core)
- Virtual Reality (non-existing environment)
- Hardware Simulation (nuclear reactor core)

Entertainment

- Digital Television
- Video on-Demand
- Interactive Portals
- Interactive Games

Wireless Networking Standards

Market Name Standard	GPRS/GSM 1xRTT/CDMA	Wi-Fi™ 802.11b	Bluetooth™ 802.15.1	ZigBee™ 802.15.4
Application Focus	Wide Area Voice & Data	Web, Email, Video	Cable Replacement	Monitoring & Control
System Resources	16MB+	1MB+	250KB+	4KB - 32KB
Battery Life (days)	1-7	.5 - 5	1 - 7	100 - 1,000+
Network Size	1	32	7	255 / 65,000
Bandwidth (KB/s)	64 - 128+	11,000+	720	20 - 250
Transmission Range (meters)	1,000+	1 - 100	1 - 10+	1 - 100+
Success Metrics	Reach, Quality	Speed, Flexibility	Cost, Convenience	Reliability, Power, Cost

1G, 2G, 3G, 4G, 5G, 6G,&7G

G?

- **G ->Generation**

- **Generation of wireless phone technology**

Generation (G) refers to a change in the nature of the system, speed, technology, frequency, data capacity, latency etc

Generation of wireless phone technology: 1G

First mobile phones to be used, which was introduced in 1982 and completed in early 1990.

It was used for voice services and was based on technology called as Advanced Mobile Phone System (AMPS

Channel capacity of 30 KHz and frequency band of 824-894MHz. [5]. Its basic features are:

- **Speed-2.4 kbps**
- **Allows voice calls in 1 country**
- **Use analog signal.**
- **Poor voice quality**
- **Poor battery life**
- **Large phone size**
- **Limited capacity**
- **Poor handoff reliability**
- **Poor security**
- **Offered very low level of spectrum efficiency**

Generation of wireless phone technology: 1G

- Bad voice quality
- Poor battery, cellphones
- Big cellphones
- Better than nothing, at least its wireless and mobile
- eavesdropping by third parties

2G

- Frequency: 1800MHz (900MHz), digital telecommunication
- Bandwidth: 900MHz (25MHz)
- Characteristic: Digital
- Technology: Digital cellular, GSM
- Capacity (data rate): 64kbps

- From 1991 to 2000
- Allows txt msg service
- Signal must be strong or else weak digital signal

Why better than 1G?

Enables services such as text messages, picture messages and MMS(Multimedia message)

2.5 G

–2G cellular technology with GPRS
- **–E-Mails**
- **–Web browsing**
- **–Camera phones**

General Packet Radio Service (**GPRS**) is a packet oriented mobile data[56 to 64Kbps].

Speed : 64-144 kbps

Take a time of 6-9 mins. to download a 3 mins. MP3 song.

2.75 G : Enhanced Data rates for GSM Evolution (**EDGE**) (also known as **Enhanced GPRS** (**EGPRS**)[175Kbps

3G

- Frequency: 1.6 – 2.0 GHz
- Bandwidth: 100MHz
- Characteristic: Digital broadband, increased speed
- Technology: CDMA, UMTS, EDGE
- Capacity (data rate): 144kbps – 2Mbps

Why better than 2G?

- From 2000 to 2010
- Called smartphones
- Video calls
- Fast communication
- Mobil TV
- 3G phones rather expensive

3 G Cont.

Speed 2 Mbps

- **Increased bandwidth and data transfer rates to accommodate web-based applications and audio and video files.**
- **Provides faster communication**
- **Send/receive large email messages**
- **High speed web/more security/video conferencing/3D gaming**
- **Large capacities and broadband capabilities**
- **TV streaming/mobile TV/Phone calls**
- **To download a 3 minute MP3 song only 11 sec-1.5 mins time required.**
- **Expensive fees for 3G licenses services**
- **It was challenge to build the infrastructure for 3G**
- **Expensive 3G phones**
- **Large cell phones**

3G mobile system was called as UMTS(Universal Mobile Telecommunication System) in Europe

3.5 G

- 3.5 G grouped together dissimilar mobile telephony and data technologies
- HSDPA: High speed downlink packet access(for downloading data)
- HSUPA: High speed upload packet access.(2Mbps)
- 3.75G HSPA+: High speed packet access
- 3.9G: LTE

4G

- Frequency: 2 – 8 GHz
- Bandwidth: 100MHz
- Characteristic: High speed, all IP
- Technology: LTE, WiFi
- Capacity (data rate): 100Mbps(mobile users – 1Gbps(Stationary)
- Why better than 3G?
- From 2010 to today (2020?)
- –Mobile multimedia
- –Anytime, anywhere
- –Global mobile support
- –Integrated wireless solutions
- –Customized personal service
- Good QoS + high security

4G

LTE (Long Term Evolution) is considered as 4G technology

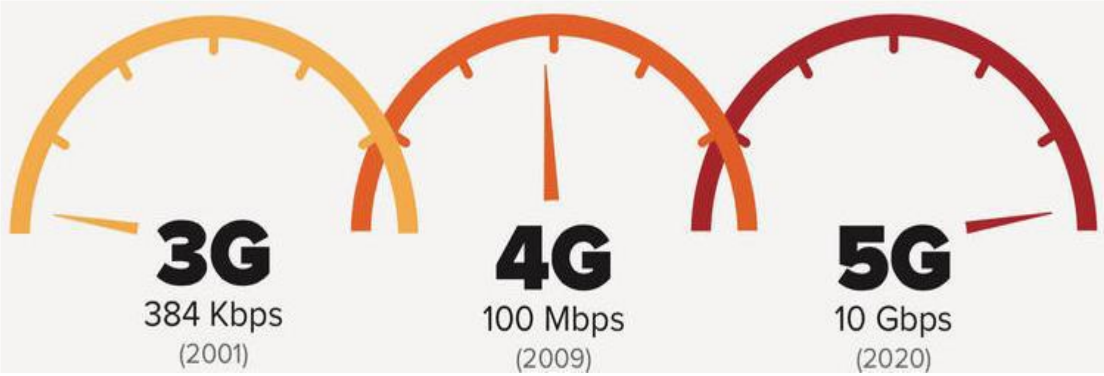
Capable of provide 10Mbps-1Gbps speed

- **High quality streaming video**
- **Combination of Wi-Fi and Wi-Max**
- **High security**
- **Provide any kind of service at any time as per user requirements anywhere**
- **Expanded multimedia services**
- **Low cost per-bit**
- **Battery uses is more**
- **Hard to implement**
- **Need complicated hardware**
- **Expensive equipment required to implement next generation network**

5G

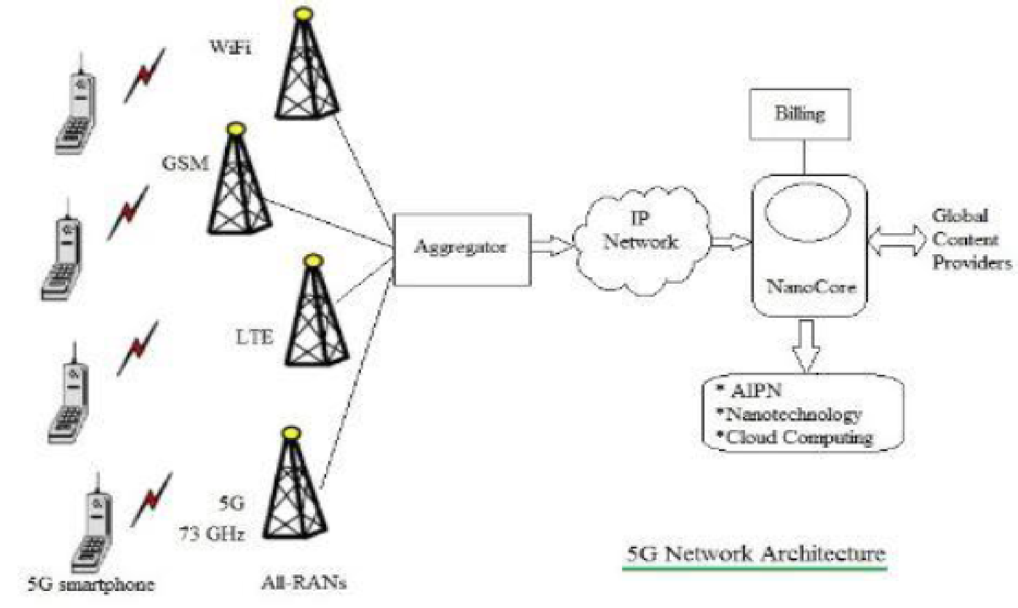
5G started from late 2010s. The main focus of 5G will be on world-Wireless World Wide Web (WWWW). It is a complete wireless communication with no limitations. The main features of 5G are :

- **It is highly supportable to WWWW (wireless World Wide Web)**
- **High speed, high capacity(**Capacity (data rate): 1Gbps – ULIMITED?)
- **Provides large broadcasting of data in Gbps.**
- **Multi-media newspapers, watch TV programs with the clarity(HD Clarity)**
- **Faster data transmission that of the previous generation**
- **Large phone memory, dialing speed, clarity in audio/video**



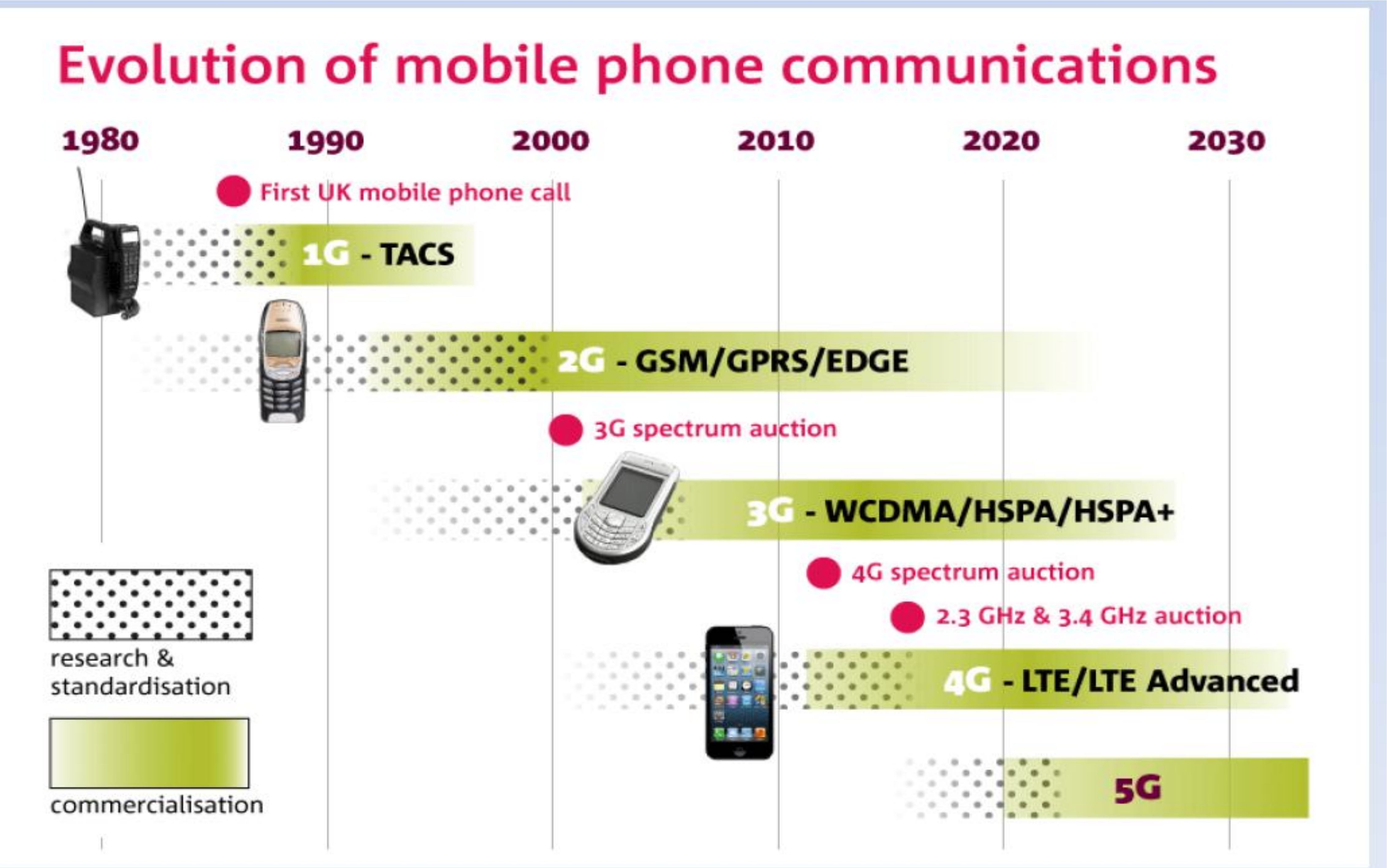
Future: 5G? 6G?

- 5G:



- 6G:
  - Integrate 5G with satellite network for global coverage
  - Ultra fast Internet access
  - Smart home/cities

- 7G:
  - Space roaming
  - World completely wireless



Evolution of mobile phone communications
1980
1990
2000
2010
2020
2030
First UK mobile phone call
1G - TACS
2G - GSM/GPRS/EDGE
3G spectrum auction
3G - WCDMA/HSPA/HSPA+
4G spectrum auction
2.3 GHz & 3.4 GHz auction
4G - LTE/LTE Advanced
5G
research & standardisation
commercialisation

Technology	1G	2G	3G	4G	5G
Start/Deployment	1970-80	1990-2004	2004-10	Now	Soon (probably by 2020)
Data Bandwidth	2Kbps	64 Kbps	2 Mbps	1 Gbps	Higher than 1 Gbps
Technology	Analog	Digital	CDMA 2000, UMTS,EDGE	Wi-Max, Wi-Fi, LTE	WWWW
Core Network	PSTN	PSTN	Packet N/W	Internet	Internet
Multiplexing	FDMA	TDMA/CDMA	CDMA	CDMA	CDMA
Switching	Circuit	Circuit,Packet	Packet	All Packet	All Packet
Primary Service	Analog Phone Calls	Digital Phone Calls and Messaging	Phone calls, Messaging, Data	All-IP Service (including Voice Messages)	High speed, High capacity and provide large broadcasting of data in Gbps
Key differentiator	Mobility	Secure, Mass adoption	Better Internet experience	Faster Broadband Internet, Lower Latency	Better coverage and no droped calls, much lower latency, Better performance
Weakness	Poor spectral efficiency, major security issue	Limited data rates, difficult to support demand for internet and e-mail	Real performance fail to match type, failure of WAP for internet access	Battery use is more, Required complicated and expensive hardware	?

Thank You

Comparison

	1G	2G	3G	4G	5G
Period	1980 – 1990	1990 – 2000	2000 – 2010	2010 – (2020)	(2020 - 2030)
Bandwidth	150/900MHz	900MHz	100MHz	100MHz	1000x BW pr unit area
Frequency	Analog signal (30 KHz)	1.8GHz (digital)	1.6 – 2.0 GHz	2 – 8 GHz	3 – 300 GHz
Data rate	2kbps	64kbps	144kbps – 2Mbps	100Mbps – 1Gbps	1Gbps <
Characteristic	First wireless communicatio n	Digital	Digital broadband, increased speed	High speed, all IP	
Technology	Analog cellular	Digital cellular (GSM)	CDMA, UMTS, EDGE	LTE, WiFi	WWWW